Thesis Update #4

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Risk Mitigation

Project Task	Risk	Justification	Mitigation
Ordering Blood Samples and IRB approval	High	• Will need EH&S bloodborne pathogen training and clearance [4] • May need Cal Poly IRB review [3][5] • Need approved storage and disposal plan for blood samples • Approval time uncertain	• Order samples early • Take EH&S Training • Submit proposal through IRB
PCB Design/Testing Iterations	High	• Need to account for design, shipping time, soldering, testing, and redesign 2 PCBs to design and test: • NIR LED + Photodiode detector PCB • ESP32 based PPG data acquisition + BGL estimation board	• Start design early • Simulate and prototype on breadboard before PCB design when practical • Break up design into parts (Ex: LED driver PCB + Photodiode front-end PCB) • Handle boards properly with ESD procedures [6]

Getting Blood Samples

- Will need to contact EH&S and IRB for more info on full timeline for training, approval, storage, and disposal
- May be more beneficial to test on the arm after IRB approval due to the additional complexities of tissue scattering, sensor placement, spacing, etc.
- Ensure NIR sensitivity for increased glucose concentration on aqueous samples
- 940 nm and 1280 nm literature-supported wavelengths for glucose detection, but 940nm cheaper and more common in research for hardware implementations

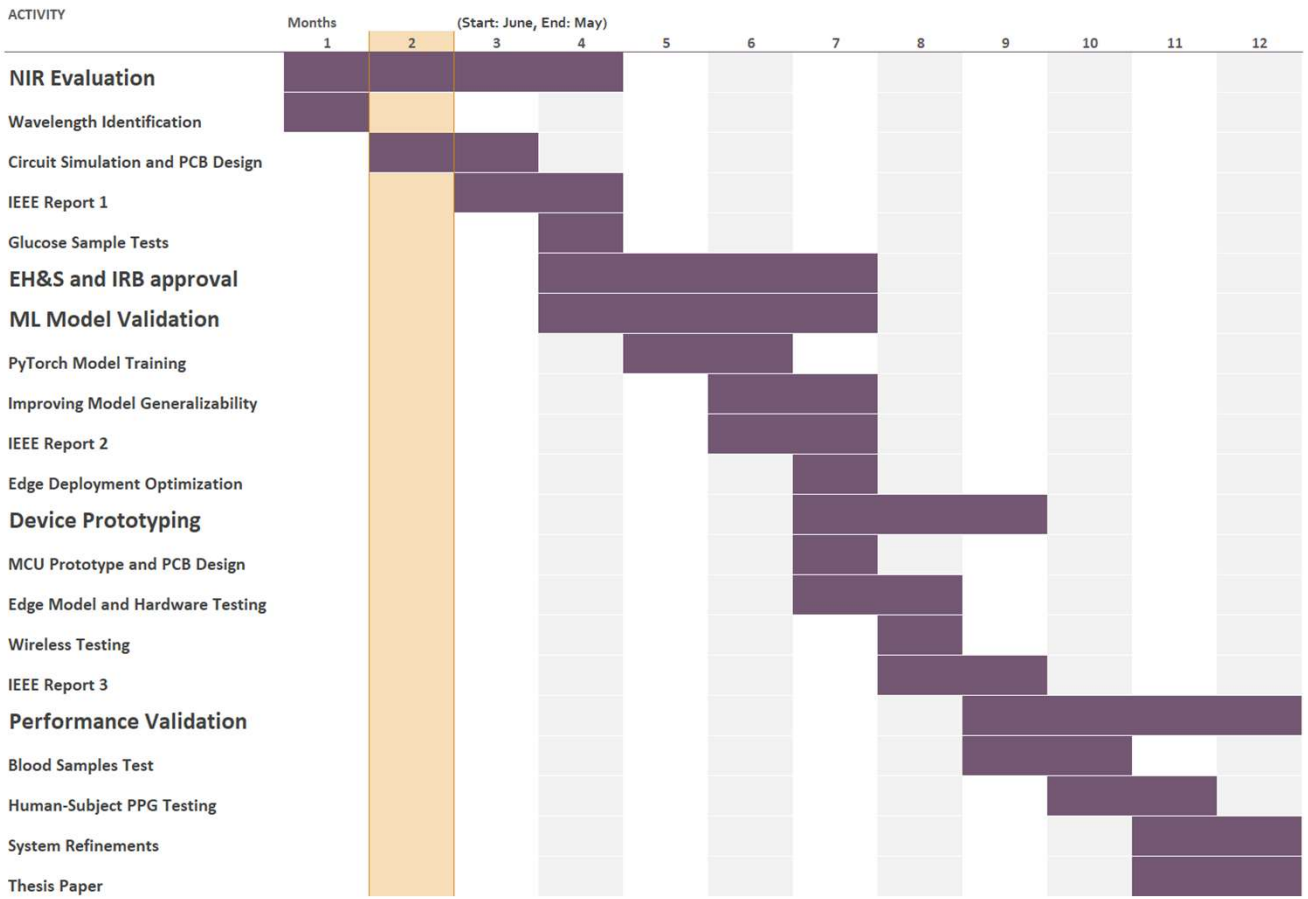
PCB Design

- Use KiCad for PCB design
- Use PCBway for manufacturing: China PCB Prototype & Fabrication Manufacturer - PCB Prototype the Easy Way
- Or Green Circuits for PCBA based in San Jose: Green Circuits: Electronic Manufacturing Services (EMS) Provider
- Hand-solder PCB for boards with less components (Ex: photodiode PCB)
- Boards with many small SMDs may require PCBA (Ex: ESP32 based glucose estimator)

ESD Safety Procedure

- Custom PCB boards present a risk of damage by ESD [6]
- Many ICs have ESD protection, but PCBs are still at risk of system-level damage [6]
- ESD damage can cause latent or immediate failures [5]
  - Harder to detect
  - Can cause days or weeks of wasted time debugging
- Follow ESD procedure presented by [5]
  - Ground surfaces to common ground
  - Ground yourself to common ground with a wrist strap
  - Discharge static charge on body and surfaces before handling electronics
- Add Transient Voltage Suppression (TVS) Diodes to PCB where needed

Thesis Timeline



Research Paper #1:
A. Hina and W. Saadeh., 2020
A noninvasive glucose monitoring SoC based on single wavelength photoplethysmography [1]

Research Paper #1

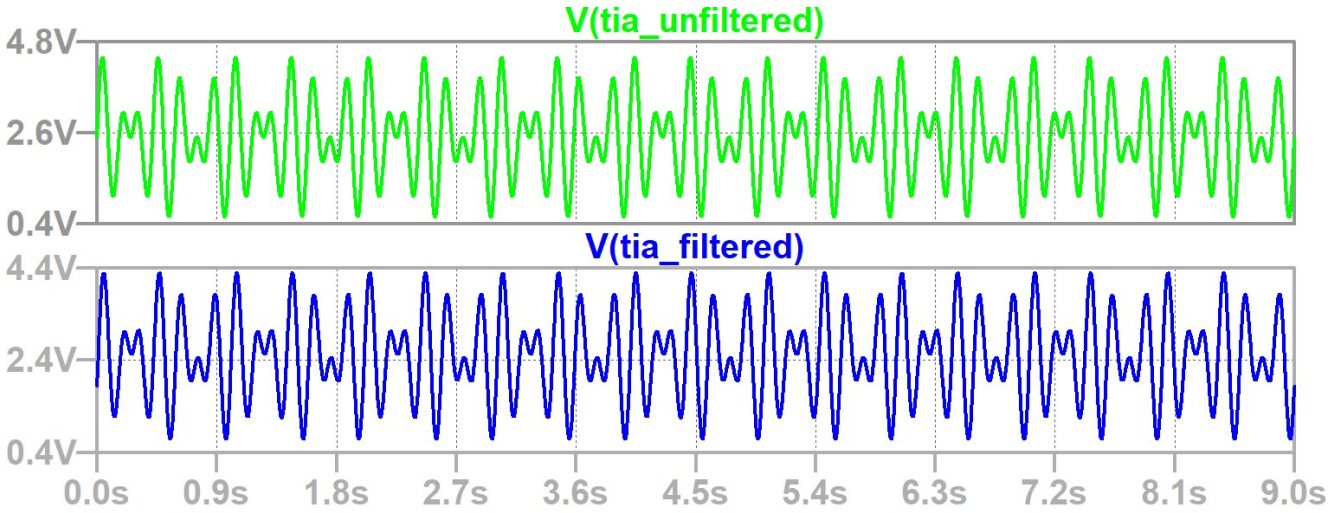
- **Hardware**
  - LED Circuit
    - 940 nm LED pulsed by clock signal to reduce power consumption, 0.8% duty cycle
    - Clock bootstrap used to step up voltage from 0.5V supply to 1.2V NIR LED forward bias
  - Photodiode
    - Produces current in response to optical power
  - Transimpedance amplifier (TIA)
    - Used an integrating op-amp to convert current to voltage
  - Low-pass filter
    - Attenuates >10Hz noise
  - Feedback DAC
    - Used to partially cancel out ambient light and DC current

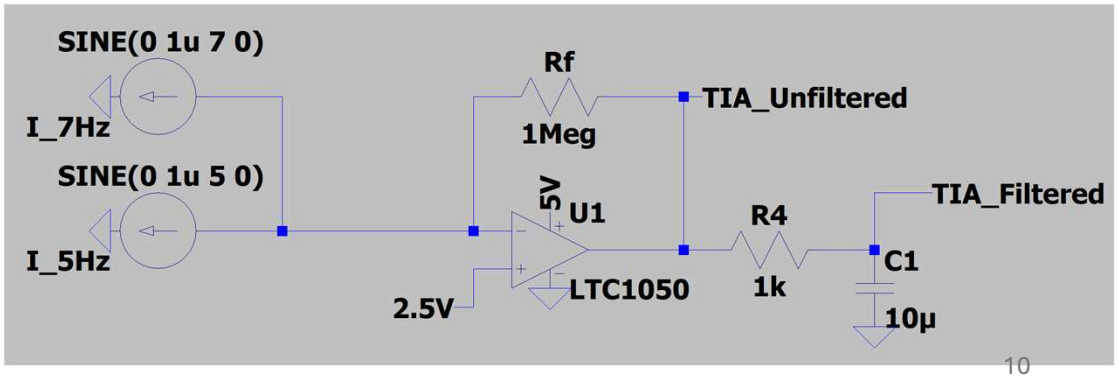
LTSpice Design

- 5V Rail and Ground
- Hardware
  - Photodiode [7]
    - Modeled as Current Source (I_{sens})
  - TIA
    - LTC1050 Op-Amp [10]
    - $R_f = 1M\Omega$
    - $V_{out} = I_{sens}R_f$
    - Ex: $I_{sens} = 10\mu A \rightarrow V_{out} = 1V$
  - Low-pass filter, ≈16 Hz [1]
    - $f_c = \frac{1}{2\pi RC}$
    - $R = 1k\Omega$
    - $C = 10\mu F$

TIA and Low Pass Filter – Noiseless Simulation

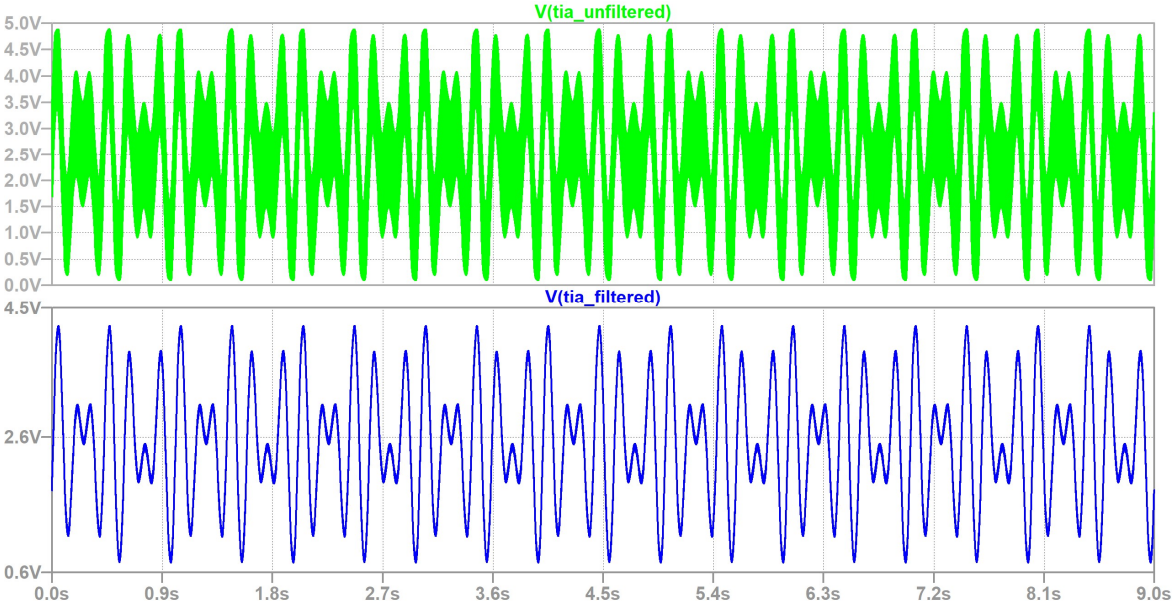
- Input Signal:
  - $I_{sens} = I_{7Hz} + I_{5Hz}$
    - I_{7Hz}, 2 μA_{p-p}
    - I_{5Hz}, 2 μA_{p-p}

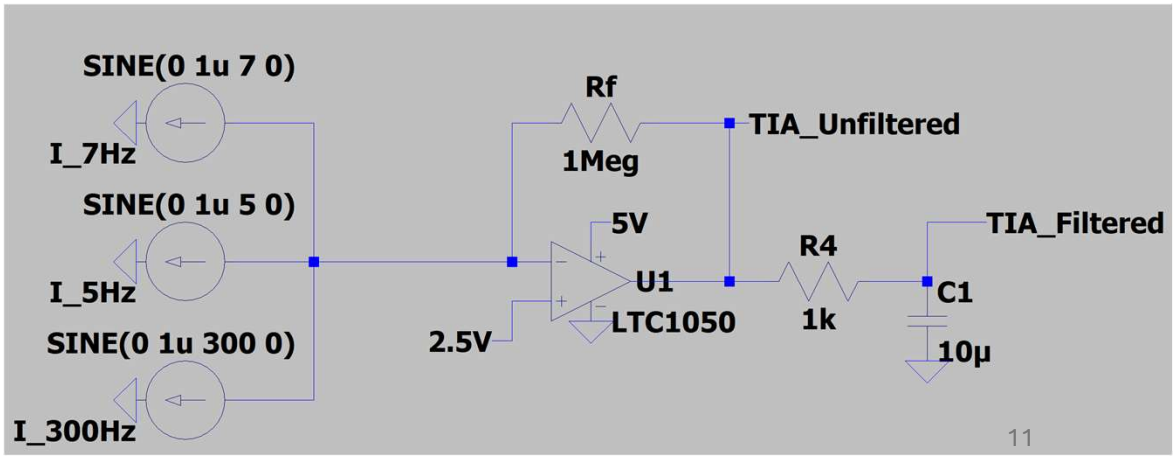




TIA and Low Pass Filter - Noisy Simulation

- Input Signal:
  - $I_{sens} = I_{7Hz} + I_{5Hz} + I_{300Hz}$
    - I_{7Hz}, 2 μA_{p-p}
    - I_{5Hz}, 2 μA_{p-p}
    - I_{300Hz}, 2 μA_{p-p}





References

[1] A. Hina and W. Saadeh, “A noninvasive glucose monitoring SoC based on single wavelength photoplethysmography,” *IEEE Transactions on Biomedical Circuits and Systems*, vol. 14, no. 3, pp. 504–515, Jun. 2020, doi: 10.1109/TBCAS.2020.2979514. Available: https://ieeexplore.ieee.org/abstract/document/9028112

[2] IRB: Human Subjects Research at Cal Poly - Office of Research - Cal Poly, San Luis Obispo

[3] Biosafety Procedure for Handling Blood Samples: bbp_template_final.pdf

[4] IRB review determination: Research Decision Chart Nov18rev.pdf

[5] EOS/ESD Association, “EOS/ESD Fundamentals Part 3: Basic ESD Control Procedures and Materials.” Available: https://www.esda.org/esd-overview/esd-fundamentals/part-3-basic-esd-control-procedures-and-materials/

[6] Texas Instruments, “System-Level ESD Protection Guide.” Available: https://www.ti.com/lit/sszb130

[7] 940 nm LED with detector: https://a.co/d/0ek5cqSH

[8]1280 nm LED: DFB laser diode at 1280nm with 100mW in TO-9 package | Innolume

[9] 1550 nm LED: LS-1550-015 Advanced Photonix | LED Emitters - Infrared, UV, Visible | DigiKey

[10] LTC1050 Op-Amp: LTC1050CN8#PBF Analog Devices Inc. | Instrumentation, Op Amps, Buffer Amps | DigiKey